

3D Creation at Your Fingertips: From Text or Image to 3D Assets

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ABSTRACT

We demonstrate an automatic 3D creation system, which can create realistic 3D assets solely from a text or image prompt without requiring any specialized 3D modeling skills. Users can either describe the object they envision in natural language or upload a reference image that records what they have seen with the phone. Our system will generate a high-quality 3D mesh that faithfully matches the users' input. We propose a coarse-to-fine framework to achieve this goal. Specifically, we first obtain a low-resolution mesh instantly by utilizing a pre-trained text/image conditional 3D generative model. Using such coarse mesh as the initialization, we further optimize a high-resolution textured 3D mesh with fine-grained appearance guidance from large-scale 2D diffusion models. Our system can create visually-pleasing results in minutes, which is significantly faster than existing methods. Meanwhile, the system ensures that the resulting 3D assets are precisely aligned with the input text or image prompt. With these advanced capabilities, our demonstration provides a streamlined and intuitive platform for users to incorporate 3D creation into their daily lives.

CCS CONCEPTS

• Information systems → Multimedia content creation.

KEYWORDS

Cross-modal Generation, Text-to-3D, Image-to-3D

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1 INTRODUCTION

3D assets have become increasingly important in a variety of applications, including video games, animation, and film. High-quality 3D assets are also essential to the construction of immersive and

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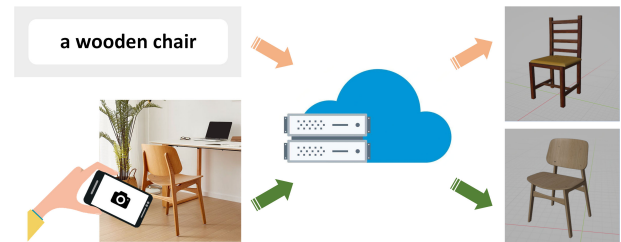


Figure 1: Demonstration of our 3D creation system to generate high-quality 3D mesh from a text or image prompt.

engaging digital worlds. However, conventional 3D asset creation is commonly time-consuming, necessitating the expertise of professional designers with significant experience in 3D software tools. Therefore, there exists a significant desire to automate the 3D asset creation process, not only to improve the efficiency of experienced 3D designers but also to democratize the process for novices.

Recent advancements in 3D generative models [3, 7] have reduced the technical barriers to creating 3D assets. When trained on a large dataset of paired 3D and text data, these models are capable of generating 3D assets conditioned on a text or image prompt in a matter of seconds. However, they struggle to create realistic 3D assets and can only generate low-resolution 3D points or meshes. Concurrently, another group of methods [5, 6, 9] leverages pre-trained large-scale diffusion models to optimize the underlying 3D representation, demonstrating their impressive capability for generating high-quality 3D assets. Nevertheless, these methods typically require hours of optimization time to produce a single sample.

In this demonstration, we integrate the advantages of the aforementioned approaches to present an automatic high-quality 3D creation system from text and image inputs, which follows a coarse-to-fine framework. In the coarse stage, we utilize a pre-trained conditional 3D generative model [3] to generate a low-resolution 3D mesh conditioned on the input text or image instantly. This mesh has accurate geometry but with worse textures. In the fine stage, we switch to refining mesh textures. We directly use the coarse mesh as the initialization and further optimize a high-resolution textured 3D mesh with fine-grained appearance guidance from large-scale 2D diffusion models [5, 6]. With such a coarse-to-fine design, our system drastically accelerates the optimization of 3D meshes while maintaining high quality. Figure 1 depicts the workflow of our system. In particular, users have the option of inputting their desired 3D object through either natural language descriptions or by uploading a reference image captured through their mobile device. Once this input is received, our system generates a high-quality 3D

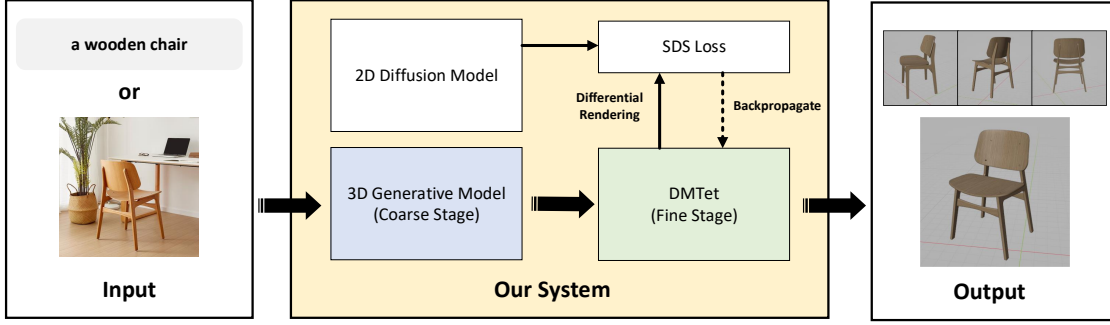


Figure 2: Overview of the coarse-to-fine text/image-to-3D generation framework in our automatic 3D creation system.

mesh that accurately matches the user’s vision and the reference image. Consequently, our system can provide users with a more streamlined and intuitive 3D creation experience. To the best of our knowledge, the work represents the first effort toward this target in the multimedia research community and has the potential to unleash users’ imagination in multimedia content creation [1, 2, 8].

2 TECHNOLOGY

Figure 2 shows the framework of our proposed system, which is composed of two components: coarse mesh generation and fine mesh optimization. The following sections provide more details.

2.1 Coarse Mesh Generation

Given a human-written natural language description or a reference image captured by the mobile device, we aim to generate a realistic 3D asset that faithfully respects the contexts present in the input text or image prompt. To achieve this goal, we first employ a state-of-the-art conditional 3D generative model, Shap-E [3], to generate 3D meshes from a text or image. In particular, Shap-E first trains an encoder to produce latent representations of 3D assets via attention mechanism widely adopted in Vision Transformers [4, 11, 12]. Then a diffusion prior model is trained on the latent representations produced by the encoder to generate 3D meshes conditioned on either images or text descriptions. The encoder and diffusion prior model are both trained on a large-scale dataset of 3D assets with corresponding renderings, point clouds, and text captions. In this component, we use the pre-trained diffusion prior model in Shap-E as our coarse mesh generation model to produce a low-resolution mesh from text or image instantly.

2.2 Fine Mesh Optimization

The generated meshes from the coarse stage typically have accurate geometry but with unrealistic textures. Therefore, in this stage, we further refine the coarse mesh through an online optimization process to make it more realistic. Inspired by [5], we adopt DMTet [10] as the 3D asset representation for the fine stage of optimization, which is parameterized as the MLP network Φ . The 3D shape is represented as a deformable tetrahedral grid (V_T, T) , where V_T is the vertices in the grid T . For each vertex $v_i \in V_T$ of the tetrahedral grid, Φ is optimized to predict the SDF value $s(v_i)$ and the deformation offset Δv_i . We use the low-resolution mesh from the coarse stage as an initialization for the mesh geometry of DMTet in this fine stage. This can alleviate the need for geometry optimization and thus drastically accelerate the optimization process.

During optimization, images were rendered from DMTet at arbitrary camera poses using a differentiable rasterizer. We then leverage large-scale pre-trained diffusion models to optimize both $s(v_i)$ and Δv_i for each vertex v_i through score distillation sampling (SDS) [9]. Specifically, when users input a text prompt, we employ a text-conditioned diffusion model known as Stable Diffusion as the supervision model for text-to-3D generation. On the other hand, when the input is a reference image, we utilize an image-conditioned diffusion model called Zero-1-to-3 [6] as the supervision model for image-to-3D generation. By doing so, SDS pushes the rendered images towards the higher-density regions under the text/image-conditioned diffusion prior, i.e., to be realistic and resemble the given input text/image prompt. When the optimization process is finished, we extract a textured 3D mesh from the optimized DMTet using the differentiable marching tetrahedra algorithm [10].

2.3 System and User Interface

Our automatic 3D creation system can be accessed through a Web browser. As shown in Figure 1, a user has the option to either type a sentence describing the desired 3D asset or upload a reference image from their mobile device. In parallel, a remote server remains on standby to receive the user’s input and execute the coarse mesh generation and fine mesh optimization processes sequentially. Finally, the created 3D mesh is returned and visualized in the Web browser. The users can download their satisfied 3D assets for their subsequent utilization in various ways. These assets can be effortlessly printed using a 3D printer, or seamlessly integrated into virtual reality applications, making 3D printing and virtual reality experiences more engaging and accessible. The system is currently run on a server with an NVIDIA A100 GPU. With the current setup, coarse mesh generation consumes around 10 seconds, and fine mesh optimization takes about 5 minutes.

3 EVALUATION

To evaluate our proposed system quantitatively, we conduct user studies by inviting 10 labelers from different educational backgrounds to annotate the generated 3D meshes. We ask each labeler to prepare 5 text & 5 image prompts and use our system to create the corresponding 3D assets. Then the labelers are asked to evaluate each generated 3D asset on a five-point ordinal scale (5: Excellent; 4: Good; 3: Neutral; 2: Bad; 1: Very Bad). Here we only treat the generated 3D assets with scores of more than 3 as the satisfying ones. According to all labelers’ feedback, the satisfying rate for text-to-3D and image-to-3D results are 76% and 70%, respectively.

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